

A speckle noise reduction technique for noncontact laser ultrasonic imaging of a rotating object

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ABSTRACT

The ultimate goal of this study is to create an ultrasonic wavefield image from a rotating object using noncontact ultrasonic excitation and sensing lasers. More specifically, this study presents a new signal processing technique to minimize speckle noise in laser Doppler vibrometer measurements. First, ultrasonic waves are generated by a noncontact Nd:Yag laser and measured using a commercial laser Doppler vibrometer from a rotating object. Next, a high-pass filter is applied to the measured ultrasonic waves to get rid of the low frequency vibrations of the rotating object. Then, the segments of the ultrasonic signal associated with a reflected light intensity lower than a threshold are removed. These steps are repeated multiple times, and all measured ultrasonic signals are averaged to improve the signal-to-noise ratio. Through this process, the speckle noise in the ultrasonic wave measurement is effectively reduced. Finally, the measurement is performed over the entire target scan area to create ultrasonic wavefield images. Experimental results obtained from a rotating disk are presented to demonstrate the effectiveness of the proposed technique.

INTRODUCTION

A laser Doppler vibrometer (LDV) measures velocity of a vibrating object by aiming an incident laser to the target point and detecting the Doppler frequency shift of the laser beam reflected from the target with respect to the incident laser beam. The reflected light is collected and then mixed with the reference beam producing an optical beating on the LDV's photodetector. This optical beating produces an electrical signal proportional to the velocity of the target point in the direction of the incident laser beam. If the target surface is rough in comparison with the laser wavelength, a continuous random distribution of light amplitude and phase, seen as bright or dark speckles, so-called speckle pattern, is formed and observed on the photodetector (Rothberg 2006). The target movements such as tilting and in-plane translation produce a change of speckle pattern, *i.e.* a sudden change in the amplitude or the phase of the measured signal (Martin and Rothberg 2009). The speckle noise, which is the main concern in this study, is proportional to the rate of the speckle pattern change. Therefore, when the target rotates at high velocity, the speckle noise level increases and contaminates the true ultrasonic signal.

Several signal processing techniques and experimental systems have been proposed to reduce the speckle noise. Vass *et al.* (2008) presents a statistical technique for detecting the presence of speckle noise in LDV signals by quantifying the amount of random impulses generated by the speckle noise. The signal sample which deviates significantly from the majority of samples in the signal is diagnosed to be distorted by speckle noise. Denman *et al.* (1996) investigated an optimum target-detector separation for a rotating target so that the speckle noise can be minimized. Pouet *et al.* (2011) introduced a multi-detector laser ultrasonic interferometer to efficiently process the speckled light reflected from rough surfaces. This new interferometer is reported to increase the interferometer sensitivity and reduce speckle noise. However, this system has a rather short stand-off distance (100 mm ~ 500 mm) compared to a conventional LDV, which has a much longer operation range.

This study reduces speckle noise by simultaneously measuring the reflected light intensity along with the ultrasonic waves. Because the speckle noise results from the change of the speckle pattern, *i.e.* the fluctuation of the reflected light intensity and phase, the speckle noise is minimized by eliminating the portion of the measured ultrasonic wave corresponding to a reflected light intensity below a certain threshold. The applicability of the proposed technique is verified by constructing laser ultrasonic propagation images from a rotating disk.

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A SPECKLE NOISE REDUCTION TECHNIQUE

The proposed speckle noise reduction technique consists of four steps: (1) The ultrasonic waves and reflected light intensity measured by the LDV are synchronously collected; (2) A high-pass filter is applied to the measured ultrasonic waves to remove low frequency mechanical vibration; (3) When the intensity of the reflected light become lower than a certain threshold, the corresponding ultrasonic signal is removed; (4) The previous steps are repeated multiple times, and all processed ultrasonic signals are averaged to improve the signal-to-noise ratio.

EXPERIMENTAL SETUP

Ultrasonic images are constructed from a circular aluminum disk with a 30 cm diameter and a 3 mm thickness, as show in Fig. 1.

The rotation speed of the disk is controlled by a speed control motor device and kept constant during the ultrasonic imaging. The ultrasonic waves are generated over a target area by scanning an Nd:Yag laser, which has a 20 Hz repetition rate, a 20 mJ peak energy and an 8 ns pulse-duration. Then, ultrasonic waves are measured at a single fixed point using a LDV. The sensing point is 12 cm off from the disk center, and the bottom of the 4×4 cm² excitation scanning area is 3 cm off from the center as shown in Fig. 2a. Ultrasonic waves are excited from a total of 841 points within the scan area, resulting in 1.4 mm grid spacing. Each ultrasonic wave and scattered light intensity signals are synchronously collected using an NI PXI-5122 digitizer with a 5 MHz sampling rate over 200 μ s.

A high-pass filter with a low cutoff frequency of 80 kHz is applied to each measured ultrasonic signal. When the light intensity of the reflected light is less than 90% of the maximum light intensity, the corresponding ultrasonic wave segment is removed. These processes are repeated 50 times at each grid point and then averaged. The measurements are performed over the entire excitation scan area to create ultrasonic wavefield images.

EXPERIMENTAL RESULTS

Ultrasonic measurement from a single point excitation and single point sensing scheme is performed first as a sanity check. The excitation and sensing points are fixed at 6 cm and 12 cm off from the disk center as shown in Fig. 2b. The rotation speed of the disk is maintained at 30 rpm during the ultrasonic measurement. Fig. 3 shows the ultrasonic signal (above) collected synchronously with the reflected light intensity (below). When the reflected light intensity drops, a significant deterioration of the ultrasonic signal is observed and this portion of the signal is removed before averaging. Fig. 4 shows the averaged ultrasonic signals with (Fig. 4a) and without (Fig. 4b) the proposed technique. It is observed that the ultrasonic signal treated with the proposed technique clearly captures pulse laser generated ultrasonic waves. The effect of the rotation speed on the speckle noise is also investigated. The ultrasonic measurement is performed at varying rotation speeds, 0 rpm, 60 rpm, and 120 rpm respectively as shown in Fig. 5. The higher the rotation velocity is, the more deteriorated the ultrasonic signal is due to speckle noise. However, even at 120 rpm, the peaks of ultrasonic waves are still properly observed.

Next, Fig. 6, Fig. 7 and Fig. 8 show the ultrasonic wavefield images constructed from the previously shown 4×4 cm² scan area with the rotating speed of 0 rpm, 60 rpm and 120 rpm, respectively. In all cases, the ultrasonic wave propagations were properly visualized.

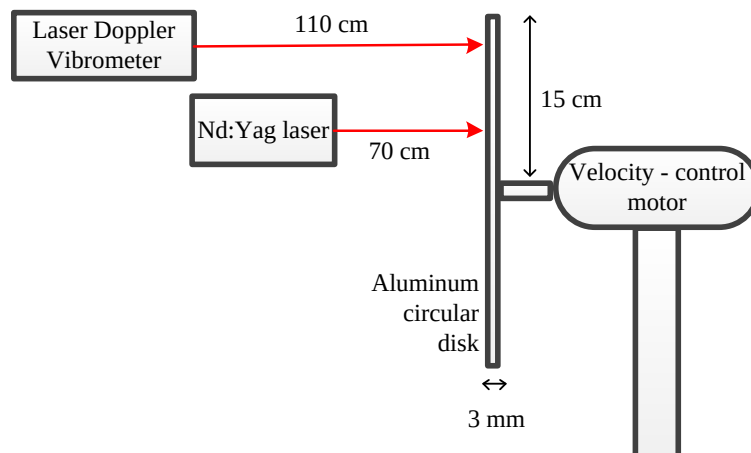


Fig. 1. Rotating disk specimen (not in scale)

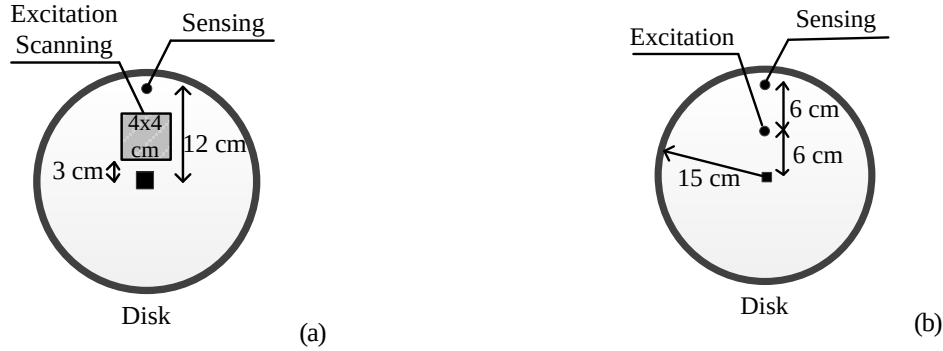


Fig. 2. (a) Multipoint excitation and a single point sensing (b) A single point excitation and a single point sensing

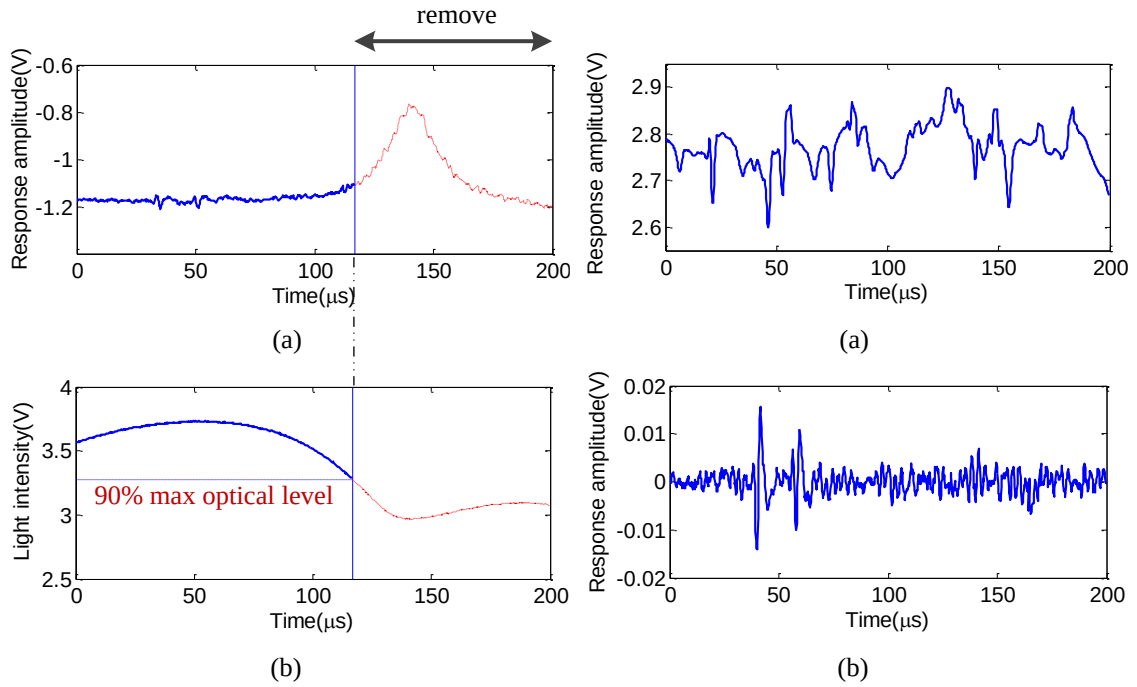


Fig. 3. Elimination of the speckle noise using the proposed technique (a) measured ultrasonic signal (b) reflected light intensity signal ultrasonic signal

Fig. 4. Comparison ultrasonic signals obtained with and without the proposed technique (a) a raw signal obtained simply by averaging (b) a processed signal obtain after applying the speckle noise reduction technique

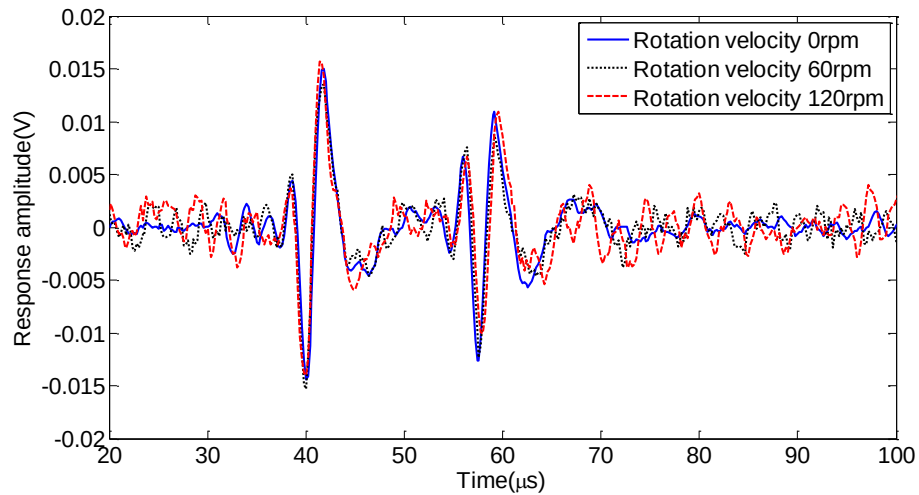


Fig. 5: Laser ultrasonic signals at different rotation velocities (applied speckle noise reduction technique)

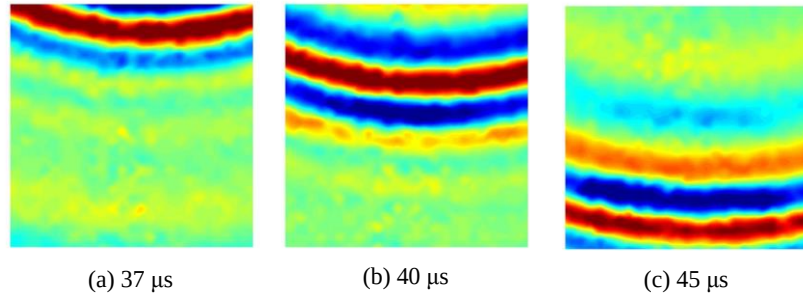


Fig. 6: Ultrasonic wave propagation image (0 rpm rotation speed)

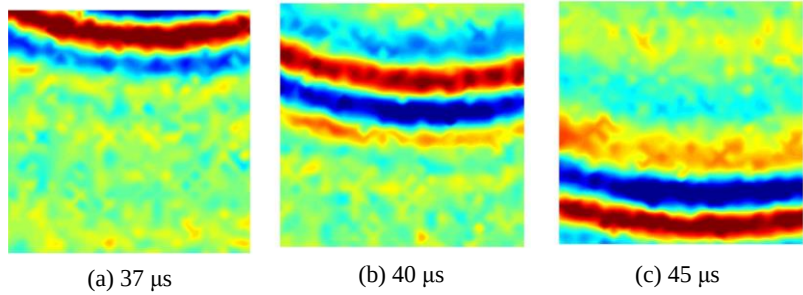


Fig. 7: Ultrasonic wave propagation image (60 rpm rotation speed)

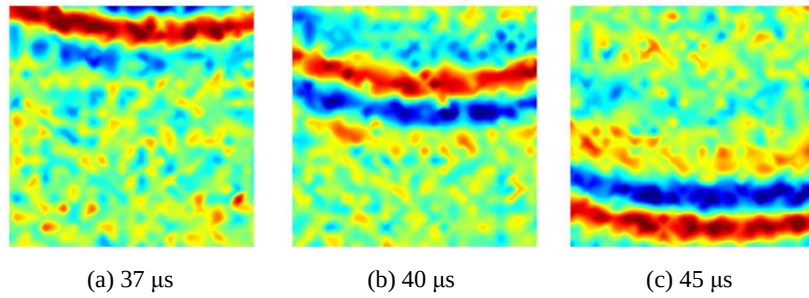


Fig. 8: Ultrasonic wave propagation image (120 rpm rotation speed)

CONCLUSION

In this study, a new speckle noise reduction technique is developed to remove unreliable portions of ultrasonic signals measured by a LDV. The intensity of the reflected light is measured and synchronized with the measured ultrasonic waves, and the portions of the measured ultrasonic waves corresponding to low light intensity are simply removed. It is demonstrated that the proposed technique can effectively eliminate speckle noise in LDV measured ultrasonic signals. Finally, the proposed technique is successfully used to construct ultrasonic wavefield images from a rotating disk using excitation and sensing lasers.

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